

3902 Project Math

Saurav Poudyel

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1 Character and Cannon Rotation

1.1 Character Kinematics

The character's motion is represented using a velocity vector $\mathbf{v}$ in a 2D plane. Given an orientation angle θ , the velocity is computed as:

$$\mathbf{v} = v [\sin(\theta) \quad -\cos(\theta)] \quad (1)$$

where v is the speed. The new position $\mathbf{p}_{t+1}$ at time step $t+1$ is given by:

$$\mathbf{p}_{t+1} = \mathbf{p}_t + \mathbf{v}dt \quad (2)$$

where dt is the frame duration.

1.2 Rotational Transformations and Bounds

To determine the rotated bounding box, we define four local-space corner points for the character:

$$\mathbf{p}_1 = [-w/2 \quad -h/2], \quad \mathbf{p}_2 = [w/2 \quad -h/2], \quad \mathbf{p}_3 = [w/2 \quad h/2], \quad \mathbf{p}_4 = [-w/2 \quad h/2]$$

The transformation matrix $R(\theta)$ rotates each corner:

$$R(\theta) = [\cos(\theta) \quad -\sin(\theta) \quad \sin(\theta) \quad \cos(\theta)] \quad (3)$$

Thus, the global positions of the corners are:

$$\mathbf{p}_i^{\text{rot}} = R(\theta)\mathbf{p}_i + \mathbf{p}_{\text{center}} \quad (4)$$

1.3 Cannon Rotation and Tip Position

The cannon rotates independently with an angle ϕ . The tip position is computed using:

$$\mathbf{p}_{\text{tip}} = \mathbf{p}_{\text{character}} + R(\phi)\mathbf{d}_{\text{tip}} \quad (5)$$

where $\mathbf{d}_{\text{tip}}$ is the local offset.

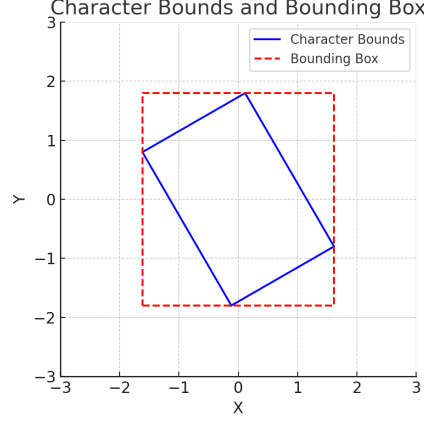


Figure 1: Rotated character bounds and its bounding box

2 Ray Tracing and Collision Detection

2.1 Raycasting Fundamentals

A ray is defined as:

$$\mathbf{r}(t) = \mathbf{o} + t\mathbf{d}, \quad t \geq 0 \quad (6)$$

where $\mathbf{o}$ is the origin and $\mathbf{d}$ is the normalized direction vector.

2.2 Ray-AABB Intersection

For an axis-aligned bounding box (AABB) defined by min/max coordinates $[x_{\min}, y_{\min}]$ and $[x_{\max}, y_{\max}]$, the intersection times are computed as:

$$t_{\min} = \max\left(\frac{x_{\min} - o_x}{d_x}, \frac{y_{\min} - o_y}{d_y}\right) \quad (7)$$

$$t_{\max} = \min\left(\frac{x_{\max} - o_x}{d_x}, \frac{y_{\max} - o_y}{d_y}\right) \quad (8)$$

If $t_{\min} \leq t_{\max}$ and $t_{\max} \geq 0$, the ray intersects the AABB.

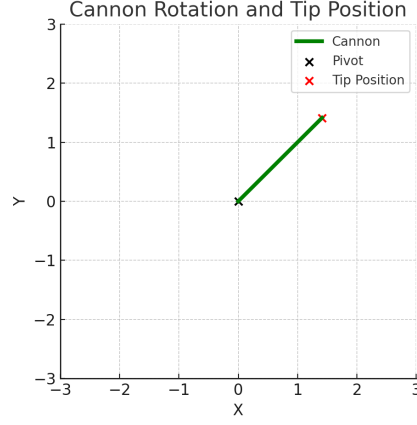


Figure 2: Cannon rotation and its effect on the tip position

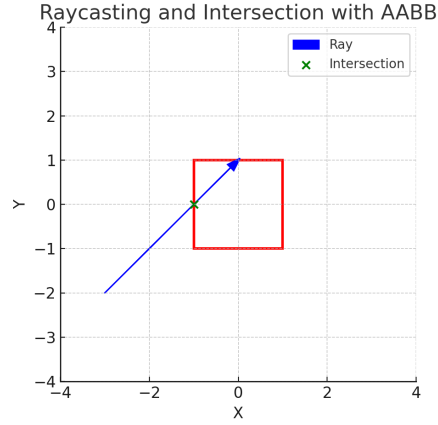


Figure 3: Raycasting and intersection with an AABB

2.3 Ray-Segment Intersection

To find the intersection of a ray with a segment $[(x_3, y_3), (x_4, y_4)]$, we solve:

$$x_1 + t(x_2 - x_1) = x_3 + u(x_4 - x_3) \quad (9)$$

$$y_1 + t(y_2 - y_1) = y_3 + u(y_4 - y_3) \quad (10)$$

for t and u , where the intersection is valid if $0 \leq t$ and $0 \leq u \leq 1$.

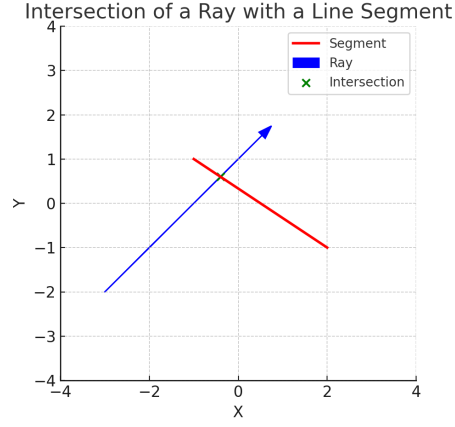


Figure 4: Intersection of a ray with a line segment

2.4 Reflection Calculation

A ray reflecting off a surface follows:

$$\mathbf{r} = \mathbf{i} - 2(\mathbf{i} \cdot \mathbf{n})\mathbf{n} \quad (11)$$

where $\mathbf{i}$ is the incident vector, $\mathbf{n}$ is the normal, and $\mathbf{r}$ is the reflected ray.

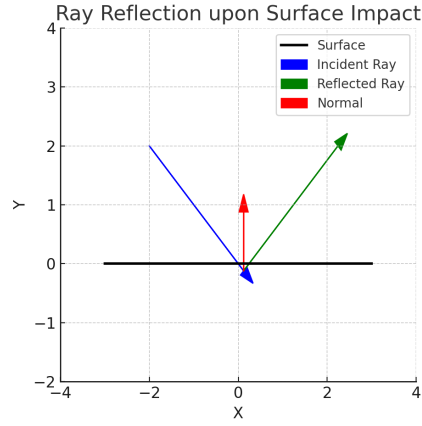


Figure 5: Reflection of a ray upon impact with a surface